

What we're doing

Evaluate **Protenix v2** at contact prediction, where a "contact" is a pyconfind side-chain contact (the same definition our `contacts_v1` training documents use). We score Protenix in **four configs** = {single-seq, msa} × {distogram, structure} on two eval sets: the FoldBench-100 monomers from exp12 (reusing its outputs) and all 454 low-MSA / structurally-novel candidates from exp65.

Ground truth & metric

Ground truth: pyconfind, `native\_only=True`, contact degree ≥ 0.001 , primary-sequence separation ≥ 6 (identical to contacts_v1). Predictors: - **distogram** — rank pairs by $P(\text{rep atoms within } 8 \text{ \AA})$ from the distogram. - **structure** — run pyconfind on the predicted structure, rank by degree.

Metric: **contacts @ L** (precision among top-L ranked pairs; also L/2, L/5), aggregate (sep ≥ 6) and split short / medium / long. Candidate pairs are restricted to GT-resolved residues, identically across configs.

Why the two predictors differ

The **structure** predictor uses pyconfind on the prediction — the *same* side-chain-contact notion as the ground truth, so it's apples-to-apples. The **distogram** predictor ranks by a $\text{CB-CB} \leq 8 \text{ \AA}$ distance notion against a side-chain-contact ground truth, so it carries an inherent representation gap. Both are what the issue asked for; expect $\text{structure} \geq \text{distogram}$.

Validation (FoldBench-100 subset)

On a 12-protein size-spanning subset: alignment identity 1.000 everywhere; MSA $\gg$ single-seq for both predictors (structure aggregate: MSA 0.76 vs SS 0.22); structure > distogram in MSA mode (0.76 vs 0.42). The one failure (8eb9_A, ~baseline across configs) is independently a Protenix failure in exp12 (msa RMSD_CA 8.8 Å), so the eval faithfully reflects model quality, not a scoring bug.

Headline result

Scored all 554 proteins (FoldBench-100 + 454 exp65), 4 configs, alignment identity ≥ 0.95 everywhere. The **structure** predictor beats the **distogram** predictor throughout (it shares pyconfind's contact notion). On FoldBench-100 (natural, deep-MSA) MSA $\gg$ single-seq (structure agg contacts @ L 0.77 vs 0.25); on exp65 (low-MSA / novel) the gap **collapses** (0.70 vs 0.58).

MSA advantage scales with MSA depth

exp65 long-range structure contacts @ L by Neff tier (MSA vs single-seq): orphan 0.41/0.42 (gap -0.01), low 0.45/0.43, marginal 0.50/0.38, deep 0.45/0.30 (gap $+0.16$). **For orphan/low-Neff proteins single-sequence is on par with MSA mode** — the regime that matters for a single-sequence model. Novel folds are the hardest stratum (long structure @ L: novel 0.39/0.28).

R-precision: ranking quality without the density cap

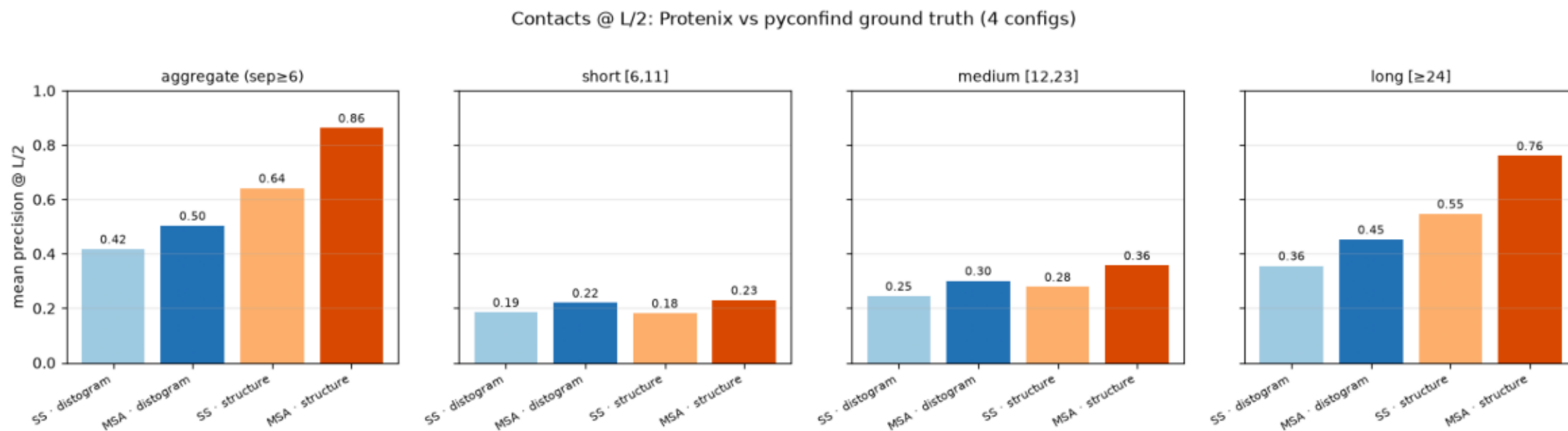
precision@L understates short-range accuracy because pyconfind contacts are sparse (there aren't L short-range contacts to find). ****R-precision**** (precision@R, R = the bin's true-contact count; ceiling 1.0 for every protein) fixes this. In R-precision the structure·MSA config is ****flat across separations**** — FoldBench short 0.84 $\approx$ medium 0.85 $\approx$ long 0.83 (vs 0.12/0.16/0.55 at precision@L) — so the model ranks short contacts just as well; the precision@L dip was a density artifact. The MSA-depth gap is cleaner too: exp65 long-range structure R-precision MSA–SS gap runs -0.01 (orphan) $\rightarrow +0.27$ (deep).

Where the data lives

`data/contact\_precision\_all.csv` (tidy scores), `plots/` (contacts @ L by config/range + the stratifications), and the HF bucket `data/protenix-contacts-eval-exp74/` (raw Protenix outputs + the full pyconfind contact tables, 696k contacts). See the README for the full reproduction spec.

contacts_at_L_2_by_config_and_range.png

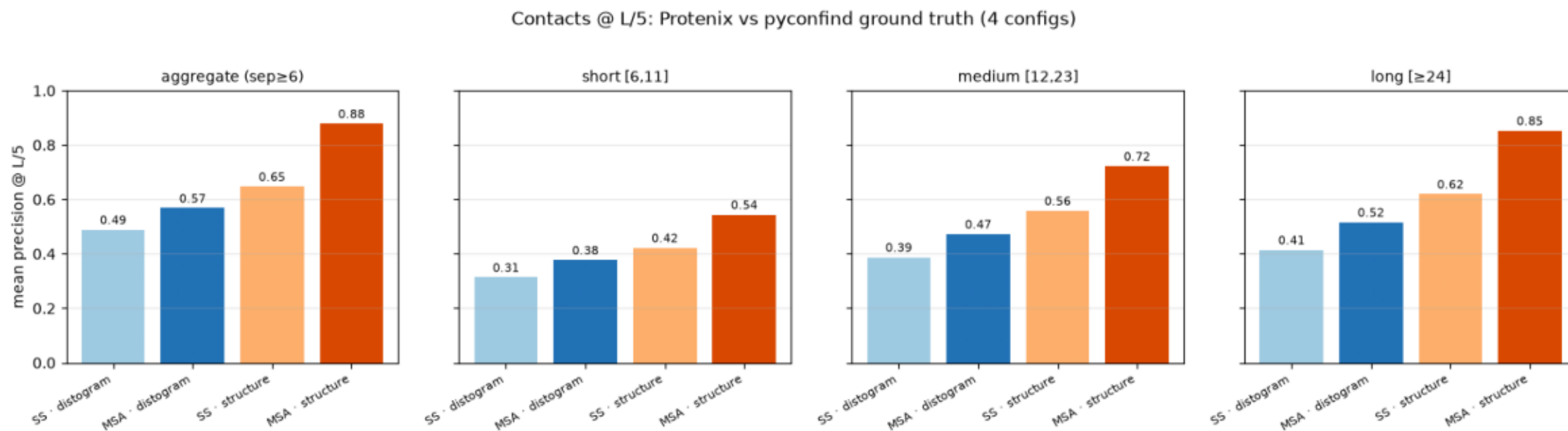
Mean precision @ L/2 vs pyconfind contacts (degree $\geq$ 0.001, sep $\geq$ 6), per config, aggregate and by range. distogram=rank by P(CB-CB $\leq$ 8Å); structure=rank by pyconfind degree on the predicted CIF.



```
$ plot.py --precision-csv data/contact_precision_all.csv --meta-csv data/contact_eval_meta_all.csv --out plots
```

contacts_at_L_5_by_config_and_range.png

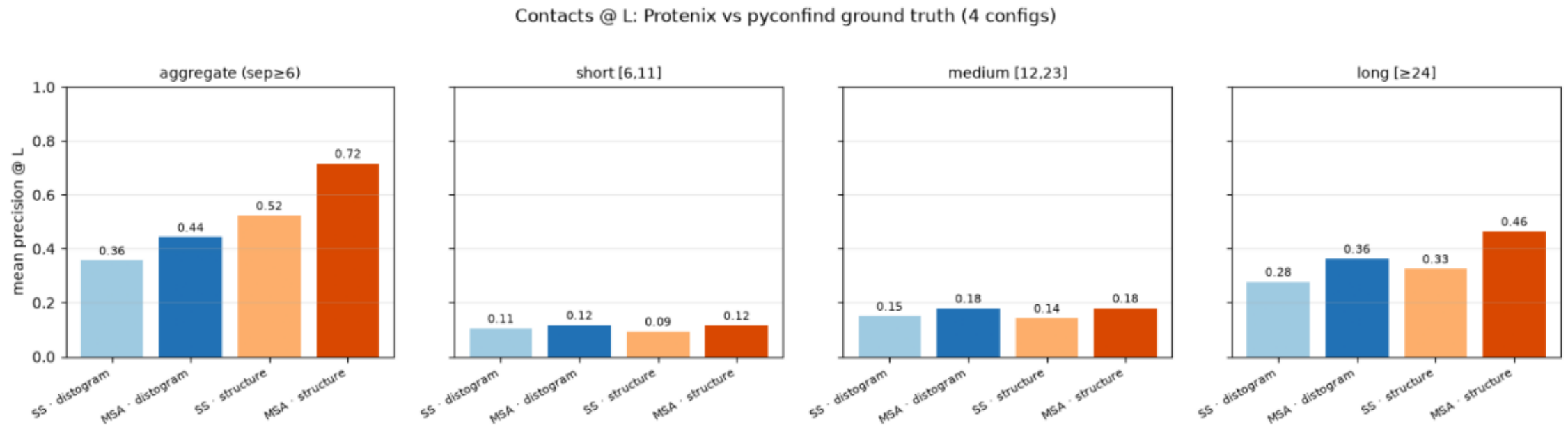
Mean precision @ L/5 vs pyconfind contacts (degree $\geq$ 0.001, sep $\geq$ 6), per config, aggregate and by range. distogram=rank by P(CB-CB $\leq$ 8Å); structure=rank by pyconfind degree on the predicted CIF.



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```

contacts_at_L_by_config_and_range.png

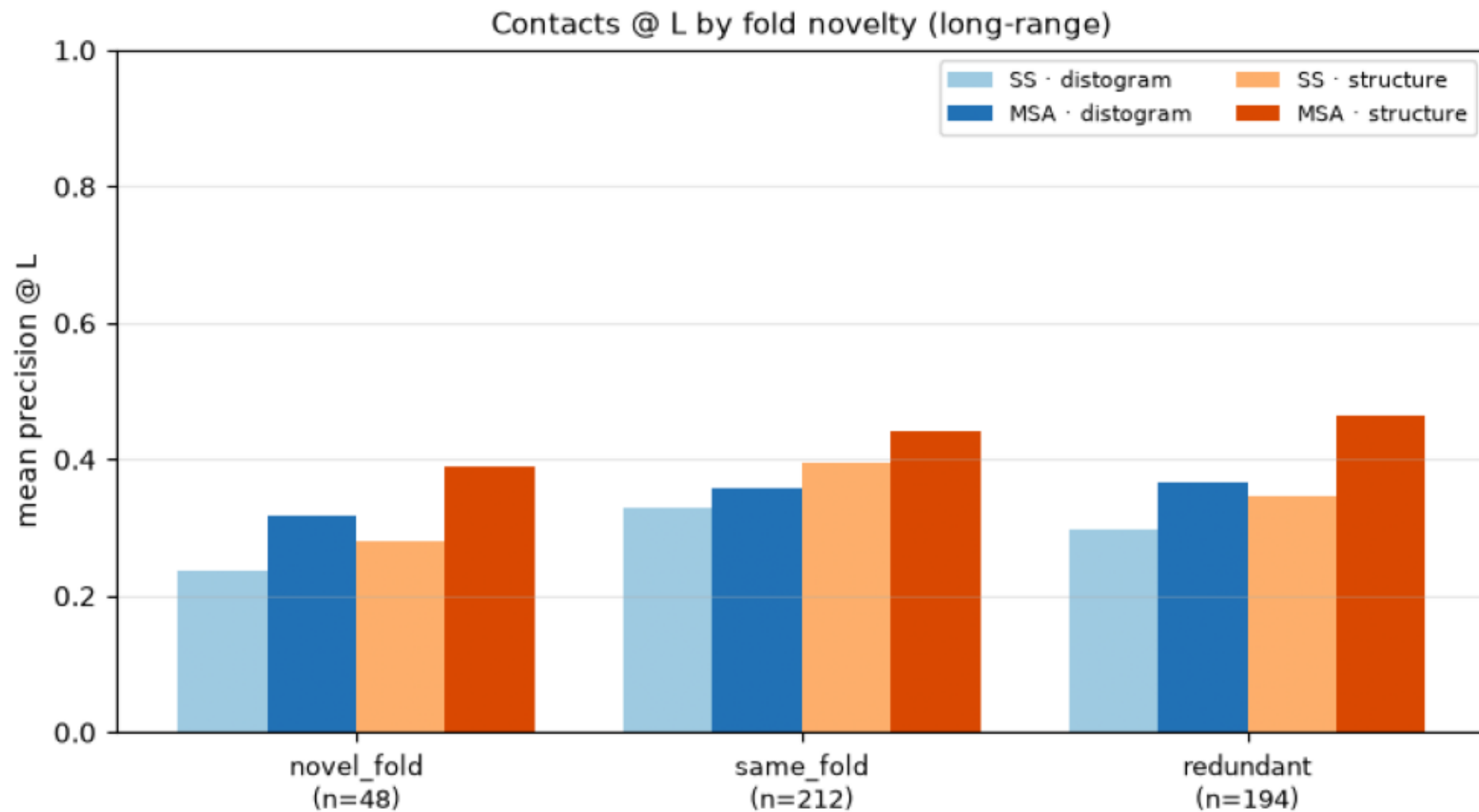
Mean precision @ L vs pyconfind contacts (degree $\geq$ 0.001, sep $\geq$ 6), per config, aggregate and by range. distogram=rank by $P(\text{CB}-\text{CB}\leq 8\text{\AA})$; structure=rank by pyconfind degree on the predicted CIF.



```
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```

contacts_at_L_by_fold_verdict.png

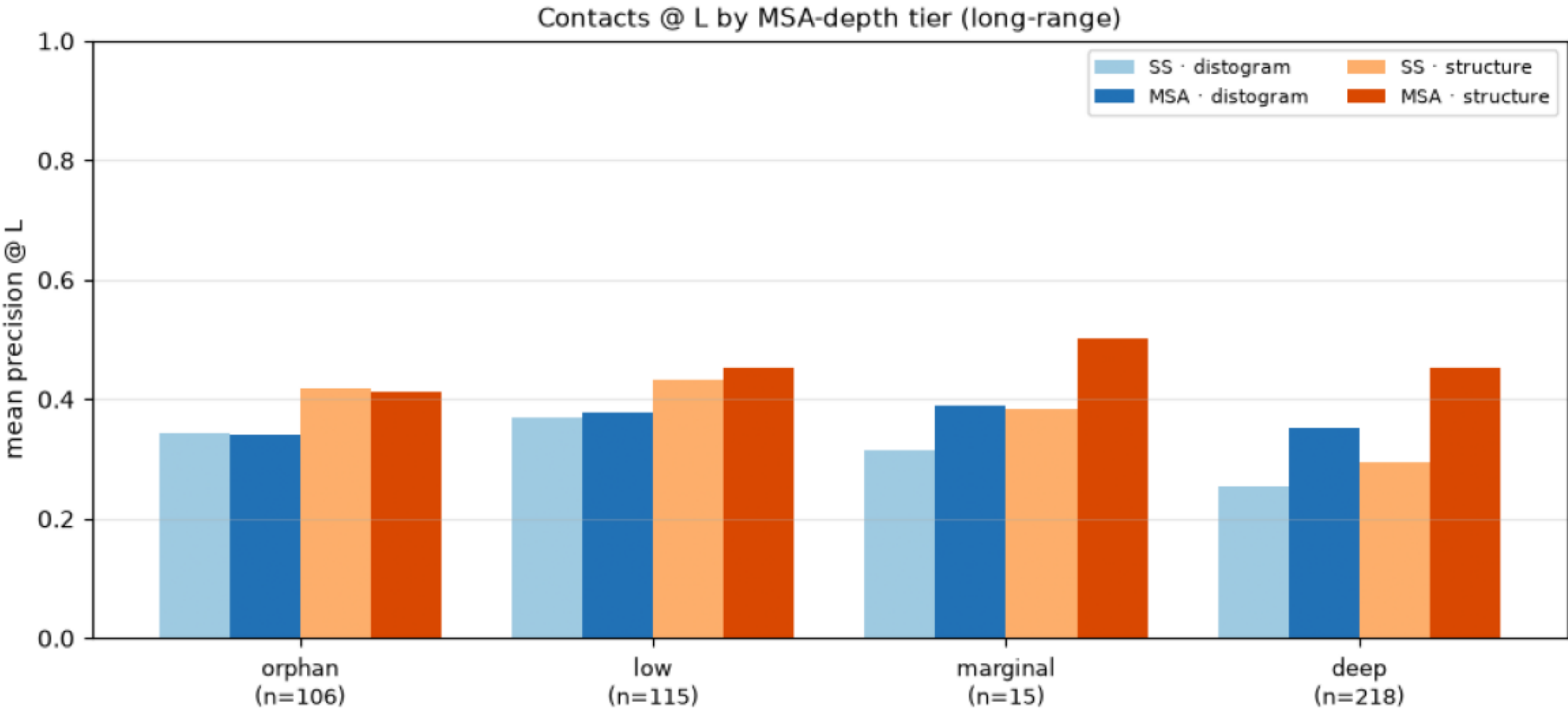
Contacts @ L by fold novelty (long-range)



```
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```

contacts_at_L_by_neff_tier.png

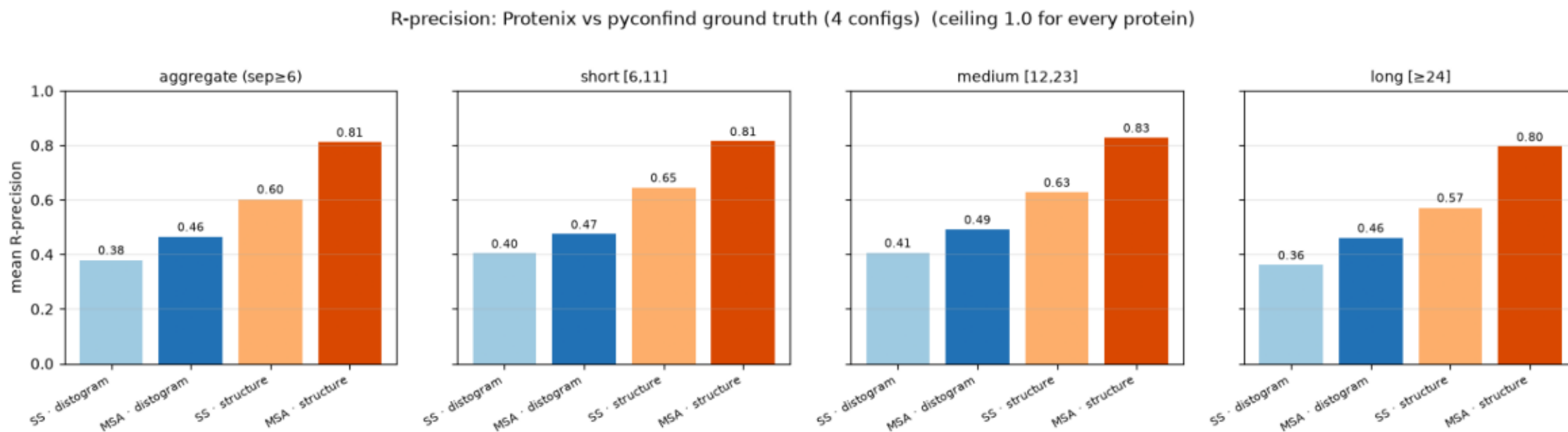
Contacts @ L by MSA-depth tier (long-range)



```
$ plot.py --precision-csv data/contact_precision_all.csv --meta-csv data/contact_eval_meta_all.csv --out plots
```

contacts_at_R_by_config_and_range.png

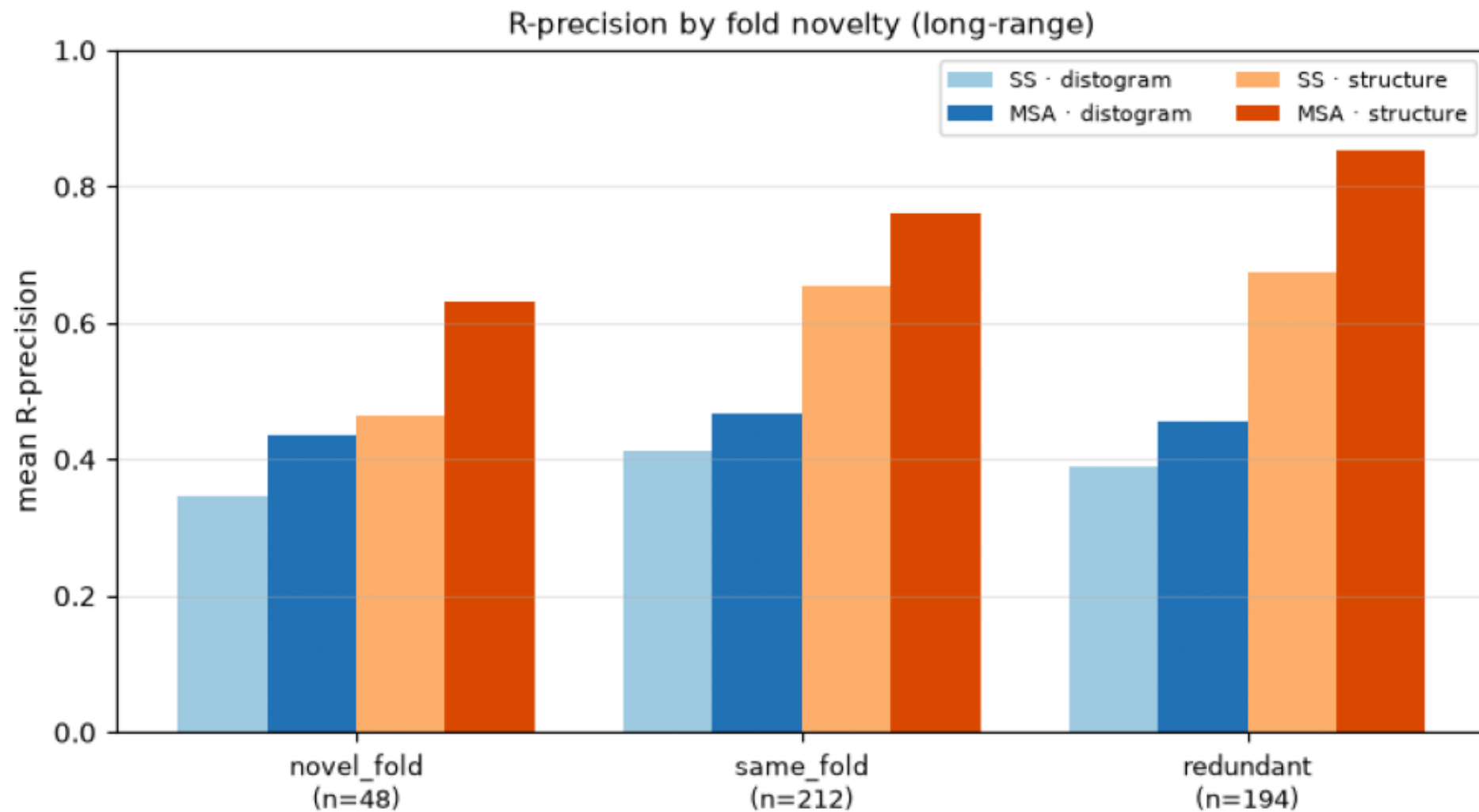
Mean R-precision vs pyconfind contacts ($\text{degree} \geq 0.001$, $\text{sep} \geq 6$), per config, aggregate and by range. distogram=rank by $P(\text{CB}-\text{CB} \leq 8\text{\AA})$; structure=rank by pyconfind degree on the predicted CIF. R-precision cutoff = #true contacts, so a perfect ranker = 1.0.



```
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contacts_at_R_by_fold_verdict.png

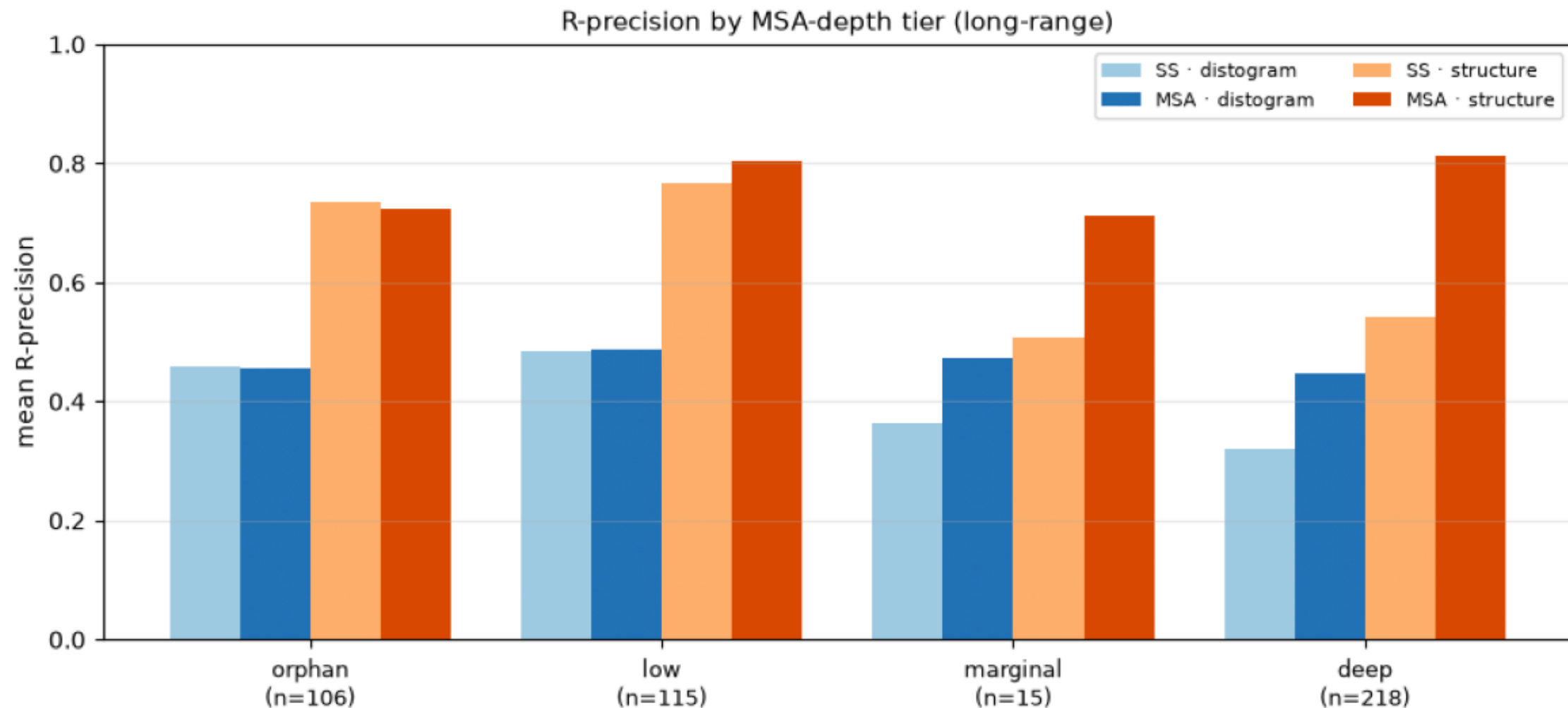
R-precision by fold novelty (long-range)



```
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```

contacts_at_R_by_neff_tier.png

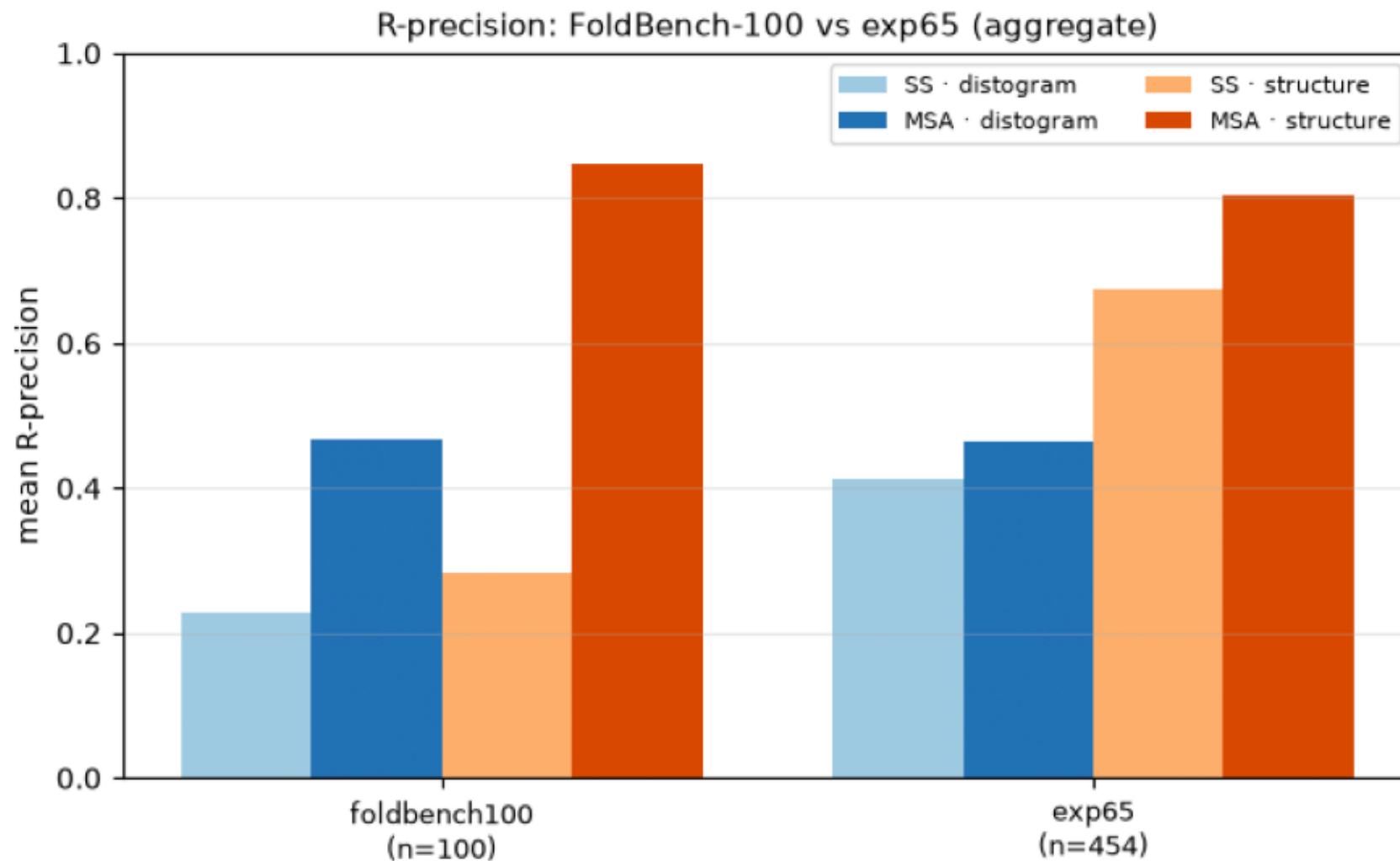
R-precision by MSA-depth tier (long-range)



```
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contacts_at_R_foldbench_vs_exp65.png

R-precision: FoldBench-100 vs exp65 (aggregate)

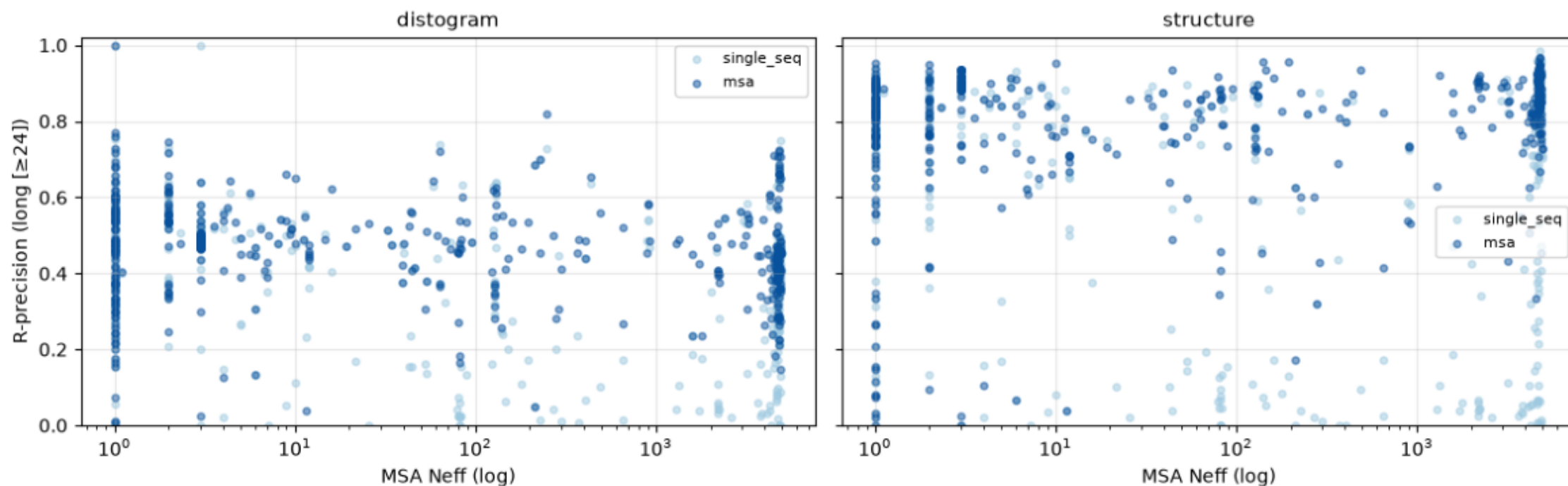


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rprecision_vs_neff_long.png

Per-protein R-precision (long range) vs MSA Neff.

R-precision vs MSA depth — long [≥ 24]



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